


January 8, 2026

Base 2, 8, 16

$$\begin{array}{cccccccccccc} & 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 & & & & & \\ \text{Binary } (11111111) & \rightarrow & (& & & & & &) & & & & & \\ & & & & & & & & & & & & & \end{array}$$
$$1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$
$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 =$$
$$= 255_{10}$$

$$\begin{array}{ccccccc} & 2 & 1 & 0 & & & \\ (377)_8 & = & 3 \cdot 8^2 & + & 7 \cdot 8^1 & + & 7 \cdot 8^0 & = & 255_{10} \end{array}$$

$$\begin{array}{ccccccc} & 1 & 0 & & & & \\ (FF)_{16} & \rightarrow & (& & & &)_{10} \end{array}$$

base 16:

$$16^{\overset{1}{F}} + 16^{\circ F} = 16 \cdot 15 + 15 = 255_{10}$$

Decimal
 $()_{10} \rightarrow ()_2$
 $()_{10} \rightarrow ()_8$
 $()_{10} \rightarrow ()_{16}$

10	A
11	B
12	C
13	D
14	E
15	F

HEX

$$\begin{array}{r|l}
 255 & 2 \\
 \hline
 254 & 127 \\
 & \hline
 & 126 \\
 & \hline
 & 63 \\
 & \hline
 & 62 \\
 & \hline
 & 31 \\
 & \hline
 & 30 \\
 & \hline
 & 15 \\
 & \hline
 & 7 \\
 & \hline
 & 6 \\
 & \hline
 & 3 \\
 & \hline
 & 2 \\
 & \hline
 & 1
 \end{array}$$

$255_{10} = 11111111_2$

A blue arrow points from the bottom of the division steps to the first remainder '1'.

114₁₀

$$\begin{array}{r|l} 114 & 2 \\ \hline 114 & 57 \\ 0 & 56 \\ & 128 \\ & 128 \\ & 14 \\ 0 & 14 \\ & 7 \\ & 6 \\ & 13 \\ & 12 \\ & 11 \end{array}$$

1110010₂

$$(0.12)_{10} \rightarrow ()_2 = (.000111)_2$$

$0.12 \times 2 = 0.24$	0	just fraction ↓
$0.24 \times 2 = 0.48$	0	
$0.48 \times 2 = 0.96$	0	
$0.96 \times 2 = 1.92$	1	
$0.92 \times 2 = 1.84$	1	
$0.84 \times 2 = 1.68$	1	

Dec.	Binary	Dec.	Binary
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111
8	1000		

A B C D E F
Hex.

Binary addition

$$\begin{array}{r}
 \times \quad 01100 \\
 \quad 10001 \\
 \hline
 \quad 11101
 \end{array}$$

$$\begin{array}{r}
 \overset{1}{+} \overset{1}{1} \overset{1}{0} \overset{1}{1} 1 0 \\
 \underline{10111} \\
 1[01101]
 \end{array}$$

$$(1)_2 + (1)_2 = (10)_2$$

$$(\overset{1}{1}0)_2 + (1)_2 = (11)_2$$

It is overflow, machine should know that it is overflow

Binary subtraction

$$\begin{array}{r}
 \overset{0}{-} \overset{0}{0} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{1}{1} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{0}{0} \\
 \hline
 1
 \end{array}
 \quad
 \begin{array}{r}
 \overset{0}{-} \overset{1}{1} \\
 \hline
 1
 \end{array}$$

$$\begin{array}{r}
 -10110 \\
 \underline{10010} \\
 00100
 \end{array}$$

$$\begin{array}{r}
 -10110 \\
 \underline{10011} \\
 00011
 \end{array}$$

1's complement

logic operation

$$(1010111)_2$$

1's complement is inverse of digits

$$(01010000)_2 \text{ 1's comp.}$$

$$(01010001)_2 \text{ 2's complement}$$

$$(0101)_2 = (5)_{10}$$

January 13, 2026

$(r-1)$'s r 's complements

radix = base

$$r=10 \quad L=4 \quad N=2468 \quad r^4=10000$$

$$r^4 - 1 = 9999$$

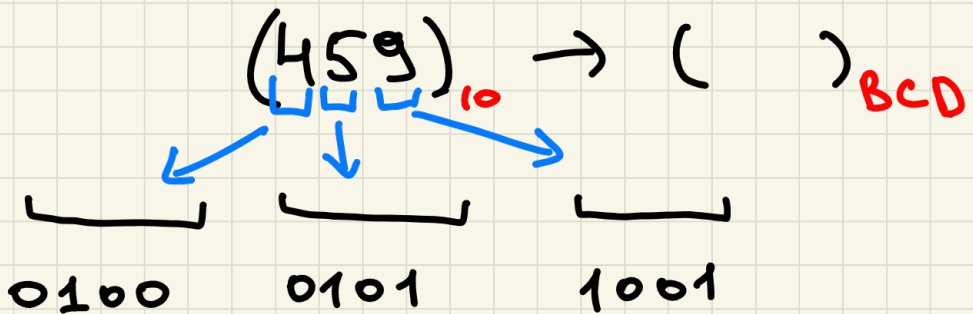
Therefore

$$\begin{array}{r} 9999 \\ - 2468 \\ \hline \underline{7531} \end{array}$$

Decimal $(185)_{10}$

Binary $(10111001)_2$

Binary Coded Decimal BCD



$$4_{10} + 8_{10} = 12_{10}$$

BCD range
(0-9)

$$0100_2 + 1000_2 = 1100_2 \rightarrow 12_{10}$$

$$0110_2 \rightarrow 6_{10}$$

$(\underline{0001} \underline{0010})_{BCD}$

$$\begin{array}{r}
 + 162 \\
 769 \\
 \hline
 931
 \end{array}$$

Binary Coded Decimal

$$\begin{array}{r}
 0001 \quad 0110 \quad 0010 \\
 0111 \quad 0110 \quad 1001 \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 1000 \quad \underline{1100} \quad \underline{1011} \\
 \text{doesn't exist in BCD} \quad \text{more than 9} \\
 \text{to solve add } 6_{10}
 \end{array}$$

$$\begin{array}{r}
 0110 \quad 0110 \\
 \hline
 \underline{1001} \quad \underline{0011} \quad \underline{0001} \\
 9 \quad 3 \quad 1
 \end{array}$$

encoding $\rightarrow$ public UTF-8
 encryption $\rightarrow$ special key

BOOLEAN LOGIC

$$A+B \rightarrow A \text{ or } B$$

$$A.B \rightarrow A \text{ and } B$$

$$A', \bar{A} \rightarrow \text{NOT } A$$

inputs

A	B	A.B	A+B	A'	B'
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0

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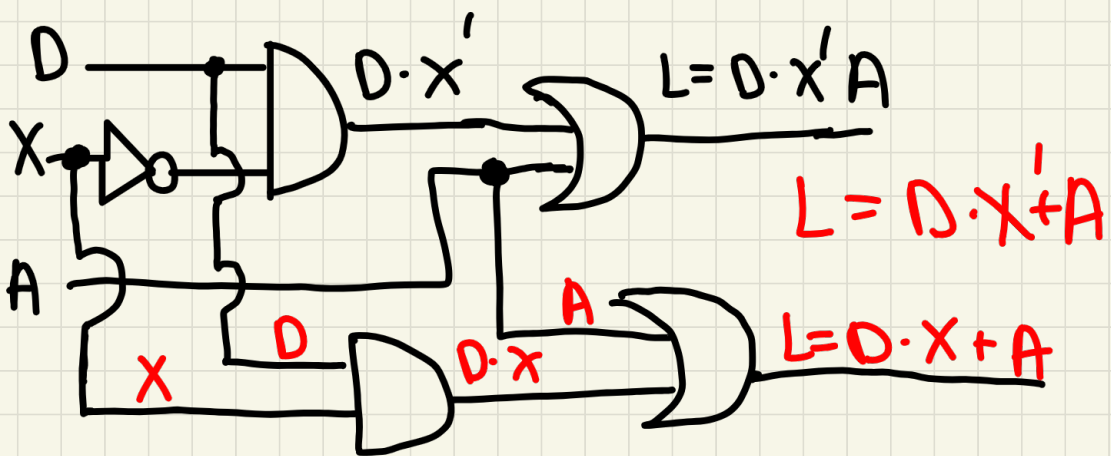
$$L = \underbrace{D \cdot X'}_{\text{inputs}} + A$$

output

D	X	A	X'	D · X'	D · X' + A
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0

output

0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1



A	B	C	AB	A'C	BC		AB + A'C
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0

1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

$$F(A, B, C) = (A+B)(A+C)$$

$$AA + AC + AB + BC \quad \text{Distributivity}$$

$$A + AC + AB + BC \quad A = AA$$

$$A(\underbrace{1 + C + B}_1) + BC$$

$$A \cdot 1 + BC = A + BC$$

Sum of Products SOP

$$F(A, B, C) = A'B'C + ABC + A'BC'$$

Product of Sum

$$\begin{aligned}
 F(A,B,C) &= (A+B)(A+C) \\
 &= (A+B) + (A+B)(A+C) \\
 &= A+B + AB+BC \\
 &= A(1+B) + BC \\
 &= A + BC = A+BC
 \end{aligned}$$

$$\begin{aligned}
 F &= A + BC + DEF \\
 &\text{Sum of Product terms} \\
 G &= (A+B)(C+D+E)(F+H) \\
 &\text{Products of Sum terms}
 \end{aligned}$$

(k)	D	X	A	X'	D X'	D X' + A	
0	0	0	0	1	0	0	1 x
1	0	0	1	0	0	1	0 x'
2	0	1	0	0	0	0	
3	0	1	1	0	0	1	
4	1	0	0	1	1	1	D' x A
5	1	0	1	0	1	1	D' x A'
6	1	1	0	0	0	0	D x A
7	1	1	1	0	0	1	

January 20, 2026

Q 1

$$y + x'z + xy' = x + y + z$$

$$y + xy' + x'z$$

$$(y+x)(y+y') + x'z$$

$$y + x + x'z$$

$$y + (x+x')(x+z)$$

$$y + x + z = x + y + z$$

Q₂

$$x'y' + x'y + xy = x'y$$

$$x'(y' + y) + xy$$

$$x' + xy$$

$$(x' + x)(x' + y)$$

$$1 \cdot x' + y = x'y$$

Q₃

$$x'y' + y'z + xz + xy + yz' \quad 13 \text{ operations}$$

$$x'y' + y'z(x + x') + xz + xy + yz'$$

$$\underline{x'y'} + xy'z + \underline{x'y'}z + xz + xy + yz'$$

$$x'y'(1 + z) + xy'z + xz + xy + yz'$$

$$x'y' + \underline{xy'z} + \underline{xz} + xy + yz'$$

$$x'y' + xz(y' + 1) + xy + yz'$$

$$x'y' + xz + xy + yz'$$

$$x'y' + xz + xy(z+z') + yz'$$

$$x'y' + xz + xyz + xyz' + yz'$$

$$x'y' + xz + xyz + yz'(x+1)$$

$$x'y' + \underline{xz} + \underline{xyz} + yz'$$

$$x'y' + xz(1+z) + yz'$$

$$x'y' + xz + yz'$$

8 operations

SOP

$$F(a,b,c) = a + bc + abc'$$

$$A(a,b,c) = \underbrace{abc}_{\text{Minterms}} + \underbrace{a'bc}_{\text{Minterms}} + \underbrace{abc'}_{\text{Minterms}} \quad \text{Canonical (product terms)}$$

K						Minterms
	a	b	c	A	A	SOP
0	0	0	0	$a'b'c'$	0	$x=1$ $x'=0$
1	0	0	1	$a'b'c$	0	
2	0	1	0	$a'bc'$	0	
3	0	1	1	$a'bc$	1	
4	1	0	0	$ab'c'$	0	
5	1	0	1	$ab'c$	0	
6	1	1	0	abc'	1	
7	1	1	1	abc	1	

$$A(a,b,c) = \sum m(3,6,7) \rightarrow \text{minterms}$$

↓
sum of products

POS (product of sum)

$$G(a,b,c) = \underline{(a+b+c')} \cdot \underline{(a'b'+c')} \cdot \underline{(a+b+c)}$$

Max terms:

$$x=0$$

POS

$$x'=1$$

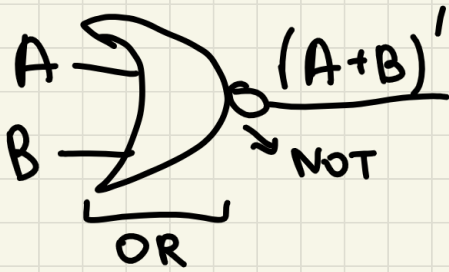
	a	b	c	P	
0	0	0	0	abc	1
1	0	0	1	abc'	1
2	0	1	0	$ab'c$	0
3	0	1	1	$ab'c'$	0
4	1	0	0	$a'b c$	0
5	1	0	1	$a'b c'$	0
6	1	1	0	$a'b'c$	0
7	1	1	1	$a'b'c'$	1

$$P(a,b,c) = \underbrace{\Pi M}_{\text{POS}} (\underbrace{0, 1, 7}_{\text{max terms}})$$

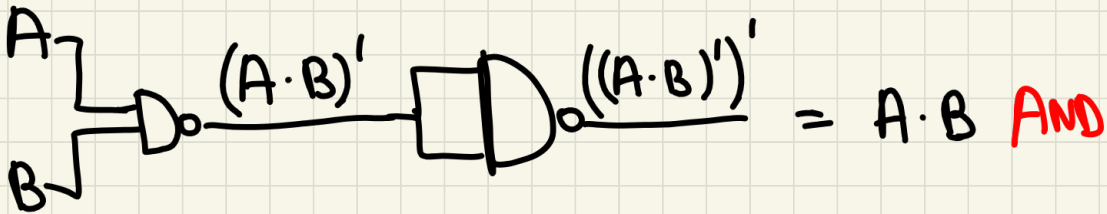
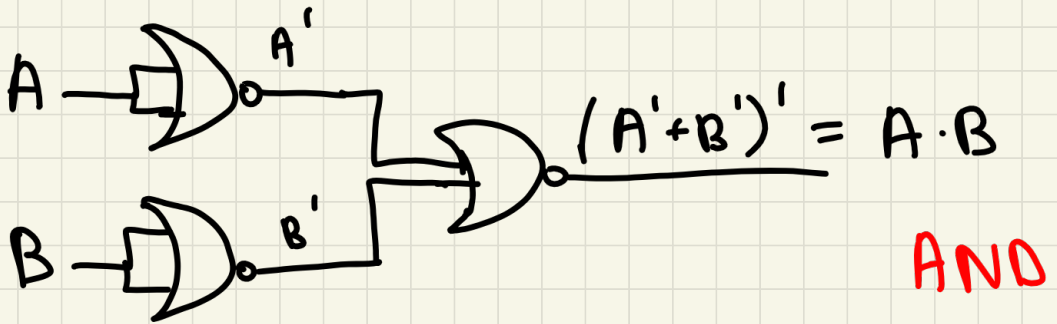
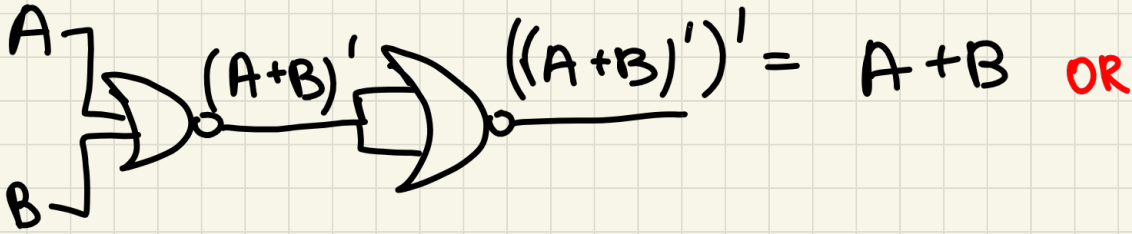
Always write Maxterms POS $x'=1$
Minterms SOP $x=1$
 $x'=0$

January 22

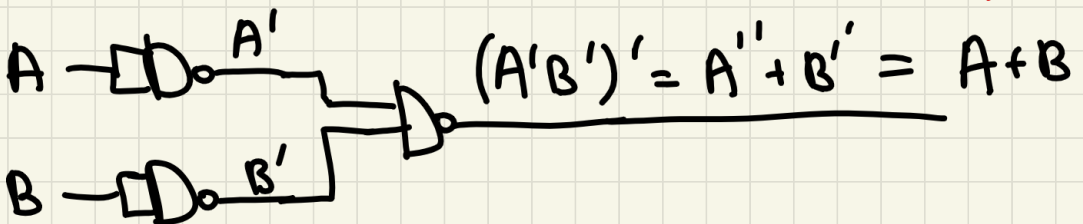
NOR Gate - Universal Gate means
Function completeness (OR, AND, NOT)



NOR



OR



$$F(a, b, c) = a'b'c + bc' + ac' \quad (\text{Standard})$$

↓
missing a
↓
missing b

$$F(a, b, c) = a'b'c + \cancel{abc'} + a'bc' + \cancel{abc'} + \underline{ab'c'}$$

$$F(a, b, c) = a'b'c + a'bc' + abc' + ab'c'$$

SOP Canonical form

$$F(a, b, c) = (a+b+c) \cdot (a+b) \cdot (a+c') \quad (\text{Standard})$$

↓
needs c
↓
needs b

$$\cancel{(a+b+c)} \cdot \underline{(a+b+c)} \cdot \cancel{(a+b+c')} \cdot \underline{(a+b+c')} \cdot (a+b'+c')$$

$$(a+b+c) (a+b+c') (a+b'+c')$$

POS Canonical form

1-Variable map

x	F(x)
0	F(0)
1	F(1)

2 variable maps

x	y	F(x)
0	0	F(0)
0	1	F(1)
1	0	F(2)
1	1	F(3)

	0	1
0	00	01
1	10	11

2 variable maps

x	y	z	F(x)
0	0	0	F(0)
0	0	1	F(1)
0	1	0	F(2)
0	1	1	F(3)
1	0	0	F(4)
1	0	1	F(5)
1	1	0	F(6)
1	1	1	F(7)

x \ yz	00	01	11	10
0	$x'z'y'$ 000 0	$x'y'z$ 001 1	$x'yz$ 011 3	$x'yz'$ 010 2
1	xyz 100 4	$xy'z$ 101 5	$x'yz$ 111 7	$x'yz'$ 110 6

→ gray codes

SOP | $x=1$
 $x'=0$

$$F(x,y,z) = \sum m(1,3,5,7)$$

$$x'y'(z+z')$$

$$x'y'(1) = x'y'$$

0	0 ₀	1 ₁	1 ₃	0 ₂
1	0 ₄	1 ₅	1 ₇	0 ₆

Group of 4

$$(2^n) = 1, 2, 4, 8, 16$$

$$x' y' z + x' y z + x y z' + x y z$$

$$z (x' y' + x' y + x y' + x y)$$

$$z (x' \underbrace{(y' + y)}_1 + x \underbrace{(y' + y)}_1)$$

$$z (\underbrace{x' + x}_1)$$

z

Shorter way to calculate

4 variable map

w x y z | f(w, x, y, z)

16 rows

w \ x \ yz	00	01	11	10
00	0000 0	0001 1	0011 3	0010 2
01	0100 4	0101 5	0111 7	0110 6
11	1100 12	1101 13	1111 15	1110 14
10	1000 8	1001 9	1011 11	1010 10

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$$F(a, b, c) = \underbrace{a \cdot b \cdot c'}_{\text{and}} + \underbrace{ab}_{\text{and}} + c$$

OR OR

x

	1	1	
	1	1	

$$SOP = x'y + z$$

② implicant

bc $b'c'$ $b'c$ bc'

a

	00	01	11	10
a' 0	0	1	1	0
a 1	1	0	1	0

$ab'c'$

$a'c$

bc

$$F(a, b, c) = \sum m(1, 3, 4, 7)$$

$$F(a, b, c) = ab'c' + a'c + bc$$

a

	bc	b'c'	b'c	bc'
a'	1	1	1	1
a	0	1	1	0

$$a'(b'c' + b'c + bc + bc') = a'$$

$$a' + c$$

wx \ yz		yz			
		y'z'	y'z	yz	yz'
wx	w'x'	1 ₀	0 ₁	0 ₃	1 ₂
	w'x	1 ₄	1 ₅ *	0 ₇	0 ₆
	wx	1 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
	wx'	1 ₈	0 ₉	1 ₁₀	1 ₁₁

Grouping annotations: Blue circles around (0,0), (0,1), (1,0), (1,1) and (1,2), (1,3), (2,0), (2,1). Green circles around (1,1), (1,3), (2,0), (2,1). Purple arrows pointing to (0,1), (1,1), (1,3), (2,0), (2,1) with labels 1, 4, 5, 3, 5 respectively.

Grouping:

groupsize 8, 4, 2, 1

$$F(w, x, y, z) = (0, 2, 4, 5, 8, 10, 11, 12, 13, 15)$$

$$F(w, x, y, z) = 1, 2, 3, 4, 5$$

1	0	0	1
1	1	0	0
1	1	1	0
1	0	1	1

Red grouping boxes: (0,0), (0,1), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2), (2,3).

January 29, 2026

February 3, 2026

$$F(A, B, C, D) = \sum m(1, 3, 4, 5, 9, 10, 11, 12, 13, 14, 15)$$

AB \ CD				
	00	01	11	10
00		1	1	
01	1	1		
11	1	1	1	1
10		1	1	1

$$F(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$$

AB \ CD				
	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

Prime : $BD, A'C'D, A'BC, ABC', ACD$

Combinational Logic Gates

Half adder

$$\begin{array}{r} A \\ + B \\ \hline \text{Sum} \end{array}$$

2 inputs : A, B

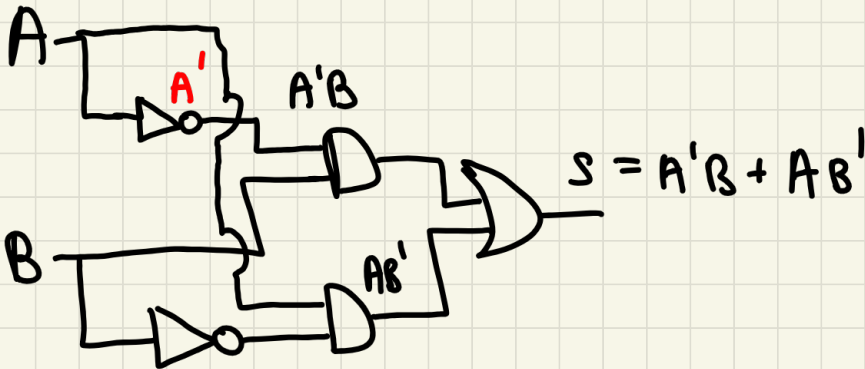
2 outputs : S, C

inputs
↔

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$S = A'B + AB'$$

$$C = AB$$



February 5, 2026

Full adder

A	B	C_{in}	S_{out}	C_{carry}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

S_{out}

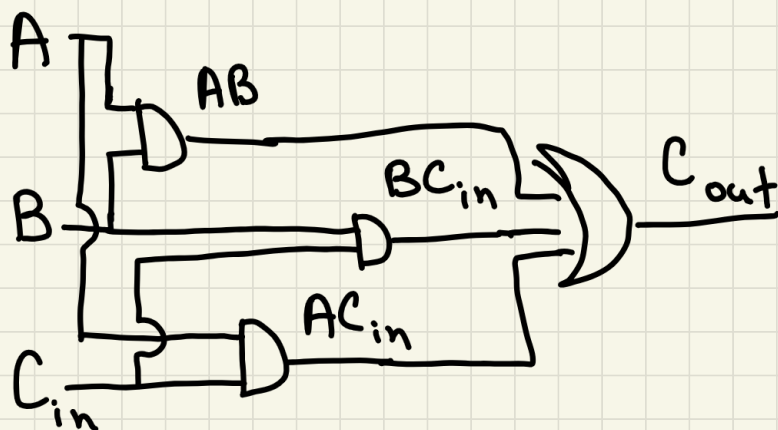
		BC			
A		00	01	11	10
		0	1	0	1
	00	0	1	0	1
	01	1	0	1	0

$$S = AB'C'_{in} + A'B'C_{in} + ABC + A'BC'_{in}$$

C_{out}

		BC			
A		00	01	11	10
	00	0 ₀	0 ₁	1 ₃	0 ₂
	01	0 ₄	1 ₅	1 ₇	1 ₆

$$C_{out} = AB + AC_{in} + BC_{in}$$



Inputs	X	Y	B_k		
	0	0	0	D	B_{k+1}
	0	0	1	0	0
	0	1	0	0	
	0	1	1		
	1	0	0		
	1	0	1		
	1	1	0		
	1	1	1		

